CHANG KOOK JEON

Platform Engineering & Cloud Infrastructure Leader

jeonck2000@gmail.com / Austin, TX / U.S. Permanent Resident, no sponsorship required
jeonck.metacog.co.kr / devops.metacog.co.kr / linkedin.com/in/metacog / github.com/jeonck

SUMMARY

Platform Engineering and Cloud Infrastructure leader with 12+ years designing, building, and operating cloud-native platforms across telecommunications, government, and enterprise environments. Proven expertise developing Kubernetes-based platforms, Internal Developer Platforms (IDP), self-service engineering capabilities, Infrastructure as Code, GitOps automation, Site Reliability Engineering, and AI-enabled infrastructure. Experienced building developer-focused platforms that accelerate software delivery through automation, reusable platform services, observability, and standardized deployment patterns. Strong background in AWS, Azure, Kubernetes, Terraform, Python automation, CI/CD modernization, and large-scale distributed systems supporting mission-critical workloads.

CORE SKILLS

Platform Engineering	Kubernetes platform engineering, platform automation, Internal Developer Platforms, engineering enablement, service reliability
Cloud Native	Kubernetes, Docker, AWS, Azure, hybrid cloud, Terraform, Infrastructure as Code, GitOps, Argo CD, distributed systems, multi-region architectures
AI & Developer Platform	LLM infrastructure, AI platform enablement, GPU infrastructure, containerized AI workloads, Jupyter platforms, FastAPI, Streamlit, AI-assisted engineering workflows
Automation & Reliability	Python automation, CI/CD pipelines, GitLab CI/CD, GitHub Actions, Jenkins, incident response, root cause analysis
Observability	Datadog, Prometheus, Grafana, OpenSearch, EFK stack, performance monitoring, reliability engineering

PUBLICATION

Reliability at Scale: Advanced SRE for Staff Engineers (2026). 354-page technical book drawn from 16 years running mission-critical telecom platforms, on the technical and organizational feedback loops behind serious reliability failures: retry storms, outages that outlive their trigger, error budgets that do not bind, and postmortems that change nothing. Fourteen chapters covering blast radius, gray failure, queueing-theory capacity, safe config delivery, incident command, and 16 public postmortems.

WORK EXPERIENCE

Cloud Infrastructure Specialist & Principal Information Systems Auditor

Nov 2025 - Apr 2026

Camino, Seoul, South Korea

- Led cloud infrastructure assessments and governance initiatives for government cloud-native environments, evaluating security posture, platform readiness, compliance, and operational risk.
- Evaluated platform readiness, security controls, operational maturity, and compliance requirements across cloud environments.
- Developed Python-based automation tools for infrastructure diagnostics, operational analysis, and platform troubleshooting.
- Supported cloud modernization initiatives through architecture reviews, automation improvements, and technical risk reduction.
- Partnered with engineering, security, and compliance teams to establish cloud governance standards and improve reliability across enterprise platforms.

DevOps Engineer & SRE Lead

Jan 2018 - Oct 2025

SK Telecom, Seoul, South Korea

- Managed enterprise-scale Azure and hybrid cloud infrastructure supporting mission-critical telecommunications platforms while maintaining 99.99% availability through SRE practices, automation, and proactive reliability engineering.
- Architected and operated Kubernetes-based cloud-native platforms supporting distributed enterprise workloads.
- Standardized Infrastructure as Code practices using Terraform, Kubernetes manifests, and automated provisioning to improve deployment consistency and scalability.
- Led modernization of seven enterprise CI/CD pipelines utilizing GitLab, GitOps methodologies, and deployment automation.
- Developed Python automation solutions that reduced manual operational effort and improved engineering productivity.
- Implemented enterprise observability using Datadog, Prometheus, Grafana, and OpenSearch to improve monitoring and incident response.
- Directed cloud optimization initiatives utilizing FinOps principles to improve resource utilization and operational efficiency.
- Partnered with engineering and product teams to improve developer experience through self-service deployment workflows and platform enhancements.

Corporate QA TF Lead & Senior Infrastructure Strategist

Jan 2014 - Dec 2017

SK Telecom, Seoul, South Korea

- Led enterprise platform readiness initiatives, including performance, stress, and endurance testing for large-scale distributed systems.
- Designed production-equivalent test environments to validate scalability, reliability, and operational readiness before deployment.
- Coordinated engineering, infrastructure, and product organizations to streamline deployment planning and improve enterprise service delivery.
- Developed operational procedures, runbooks, and validation frameworks that reduced post-deployment incidents and improved service reliability.

Infrastructure Operations Specialist, High-Availability Systems

Jan 2009 - Dec 2013

SK Telecom, Seoul, South Korea

- Managed 24x7 enterprise Linux infrastructure supporting mission-critical telecommunications platforms serving millions of subscribers.
- Led major incident response activities, troubleshooting production system failures and restoring critical services while minimizing downtime.
- Coordinated cross-functional teams during more than thirty enterprise infrastructure incidents, improving operational response and platform resilience.
- Supported business continuity, high availability, disaster recovery, and operational governance initiatives across enterprise infrastructure.

PLATFORM ENGINEERING PROJECTS

Internal Developer Platform (self-service). Designed self-service deployment workflows on Kubernetes using GitOps (Argo CD) and templated pipelines, enabling product teams to provision namespaces, pipelines, and observability without platform-team tickets.

Terraform IaC module library. Authored and published reusable Terraform modules for AWS network, compute, and Kubernetes provisioning, standardizing environment build-out and reducing new-environment setup from days to hours.

CI/CD automation pipeline. Built an end-to-end delivery pipeline with GitLab CI running self-hosted runners on Kubernetes for elastic, isolated build capacity, and Argo CD for GitOps-based continuous delivery. Containerized FastAPI and Spring Boot services using Jib to build images directly from the Gradle and Maven build, removing privileged builds from the cluster and shortening image build times.

AI/LLM infrastructure platform. Deployed GPU-backed containerized LLM and Jupyter workloads on Kubernetes with FastAPI and Streamlit service endpoints, supporting AI-assisted engineering workflows and internal developer productivity tooling.

Technical writing. Publish platform engineering, SRE, and AI infrastructure practice notes at devops.metacog.co.kr and fw-thinking.metacog.co.kr, documenting hands-on experiments and architecture decisions. Also run comparison.metacog.co.kr and comp-thinking.metacog.co.kr.

SELECTED PROJECTS

Endpoint Security Management Solution (2020 - 2021). Led end-to-end development of an enterprise-grade endpoint security management solution, establishing centralized visibility and control over diverse endpoint assets. Conducted security audits and risk assessments, applying zero-trust architecture principles to fortify the security posture.

Systems Stability and Security Assurance (2015 - 2017). Directed static analysis initiatives using Fortify to identify structural risks and implement security hardening protocols. Established a secure test bed environment validating both functional and security parameters of cross-unit interactions.

IT Stack Optimization and Best Practice TF (2014). Led a specialized task force standardizing the IT stack across OS, database, WAS, and network, authoring integrated best-practice guidelines that reduced integration-related failures.

Integrated System Maintenance and KPI Management (2009). Managed monthly KPI reporting for integrated systems, using data-driven insights to optimize maintenance cycles.

Overseas Posting, Vietnam Network Operations (2005 - 2008). Resident overseas assignment directing telecommunications network operation KPIs for Ho Chi Minh City and Hanoi, handling technology transfer and quality compliance with local operating teams.

OPEN SOURCE

Field Cases. Public knowledge system for ICT failure modes (source). A promotion test decides what gets written; notes are titled by symptom, tagged on three axes (target, layer, symptom), linked three ways, and masked while writing. Markdown in Git, published with Quartz and GitHub Actions. The notes are worked examples, not client engagements.

terraform-optimal. Terraform reference lab with guardrails at four layers: S3 remote state with native locking and CI concurrency groups, apply only from saved plan files with daily drift detection, per-environment state split, and variable validation plus OPA/conftest gates at plan time.

terraform-aws-lab. Twenty ordered AWS labs from a first S3 bucket through modules, environment separation, remote state, and CI/CD, then security scanning, state recovery, policy as code, multi-region, IAM least privilege, native tests, drift detection, ECS, and EKS.

OpsForge. One project context generates nine artifacts in sync (Dockerfile, Compose, Kubernetes manifests, Helm, Terraform, GitHub Actions, GitLab CI, Ansible, Nginx), eliminating the config drift behind many deployment incidents.

WebFusion. Shift-left DevSecOps toolkit: a headless scanner with SARIF and JSON output and CI exit-code gates, plus a local intercepting proxy with repeater and fuzzer for verification.

secbrain. Markdown, Git, and Claude second brain. A site button opens a GitHub issue that triggers Claude Code inside Actions to write and commit notes; the browser holds zero secrets and execution happens only on the runner.

Graph Engine. PageRank, HITS, betweenness, and Louvain communities validated against a degree-preserving null model with Benjamini-Hochberg correction, exported to a dependency-free D3 page at graph-engine.metacog.co.kr.

PATENTS

Intelligent Endpoint Security Management. KR20230013443A, 2023

Automated Security Information Categorization. KR20230018724A, 2023

Organizational Chart Automation for Security. KR20220035711A, 2022

Application Service Quality Indicator System. KR101567993B1, 2015

EDUCATION

MBA, Technology Management
Hanyang University, Seoul, South Korea

BS, Electronics Engineering
Kyungpook National University,
Daegu, South Korea

CERTIFICATIONS

AWS Certified Solutions Architect

Professional Engineer in Information Technology
Korean national license. Formally commissioned as an evaluation committee member on government and public-sector technology projects.

Principal Information Systems Auditor
Korean national license. Only Principal-level auditors may lead national information system audit programs, a standing conferred by the Professional Engineer license.

CCNP (Networking), RHCSA (Linux), ISTQB